

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA14

COURSE NAME: Compiler Design

COURSE OUTCOME COVERED

CO4: *Examine intermediate code generation techniques to enhance code optimization (BL4)*

Assessment Tool Weightage: 40%

QUESTIONS: 4

Total Marks:40

Annexure A

CLASS TEST

Code of Conduct:

I am M.Niharika(192365019) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:Niharika

Q. No	Understanding of Concepts (25)	Explanation (25)	Application (20)	Organization (15)	Accuracy(10)	Total Score (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
OVERALL RUBRICS VALUE						
	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 25	/ 30	/ 20	/ 15	/ 10	/ 100

Annexure A

CLASS TEST

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	15%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps

2) construct the address code

if $((a < b) \& (c < d) \text{ or } (a > d))$ then

$z = x + y * z$

else

$z = z + 1$

Given

if $((a < b) \text{ and } (c < d) \text{ or } (a > d))$ then

$z = x + y * z$

else

$z = z + 1$

expressions: $a < b$

$c < d$

$a > d$

loops: L1, L2, L3, L4, L5, L6

L1:

if $a < b$ goto L2
goto L5

L2:

if $c > d$ goto L4
goto L3

L3:

if $(a > b)$ goto L4
goto L5

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L4: $z = x + y * 2$

L4

$t1 = y * 2$

$t2 = x + t1$

$z = t2$

goto L6

L5: $z = z + 1$

L5:

$t3 = z + 1$

$z = t3$

translate $(a < b) \& (c < d) \text{ or } (c < f)$ into three address statements using Backpatching

$[(a < b) \text{ and } (c > d)] \text{ or } (c < f)$

if $(a < b)$ goto 3

goto 5

if $(c < d)$ goto 7

goto 5

if $(c < f)$ goto 7

goto 8

true

false

Construct three address code use the following production
or boolean $(a > b) \text{ or } (c < d) \text{ or } (c < f)$

$B \rightarrow B_1 \parallel B_2$

$B \rightarrow B_1 \&\& B_2$

$B \rightarrow !B_1$

$B \rightarrow E_1 \text{ rel } E_2$

$B \rightarrow \text{true}$

$B \rightarrow \text{false}$

L1: if $(a > b)$ goto Ltrue

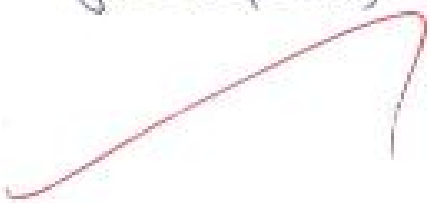
goto L2

L2: if $(c < d)$ goto Ltrue

goto L3

L3: if $(c < f)$ goto Ltrue

goto Lfalse



(4) generate the three address code for following Program fragment where $(A < C \ \& \ B < 0)$ do

if $A = 1$ then $C = C + 1$
else while $A < 0$ do
 $A = A + B$

L1:
if $A < C$ goto L2
goto LEND

L2:
if $B > 0$ goto L3
goto LEND

L3: if $A == 1$ goto L4
goto L5

L4:
 ~~$t1 = C + 1$~~
 $C = t1$
goto L1

L5: if $A < 0$ goto L6
goto L1

L6: $t2 = A + B$
 $A = t2$

goto L5

LEND:

(4) generate the three address code for following program fragment where $(A < C \ \& \ B < D)$ do

if $A=1$ then $C=C+1$
else while $A \leq 0$ do
 $A = A+B$

L1:
if $A < C$ goto L2
goto LEND

L2:
if $B > 0$ goto L3
goto LEND

L3: if $A == 1$ goto L4
goto L5

L4:
 ~~$t1 = C+1$~~
 $C = t1$
goto L1

L5: if $A \leq 0$ goto L6
goto L1

L6: $t2 = A+B$
 $A = t2$
goto L5

LEND:

(4) generate the three address code for following program fragment where $(A < C \ \& \ B < D)$ do

if $A=1$ then $c=c+1$
else while $A \neq 0$ do
 $A = A + B$

L1:
if $A < C$ goto L2
goto LEND

L2:
if $B > 0$ goto L3
goto LEND

L3: if $A = 1$ goto L4
goto L5

L4:
 ~~$t1 = C + 1$~~
 $c = c + 1$
goto L1

L5: if $A \neq 0$ goto L6
goto L1

L6: $t2 = A + B$
 $A = t2$
goto L5
LEND:

(4) generate the three address code for following Program fragment where $(A < C \ \& \ B < 0)$ do

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L1: if $A < C$ goto L2
 goto LEND

L2: if $B > 0$ goto L3
 goto LEND

L3: if $A == 1$ goto L4
 goto L5

L4: $t1 = C + 1$
 $C = t1$
 goto L1

L5: if $A < 0$ goto L6
 goto L1

L6: $t2 = A + B$
 $A = t2$
 goto L5

LEND:

(4) generate the three address code for following program fragment where $(A < C \ \& \ B < D)$ do

if $A=1$ then $C=C+1$
else while $A <= 0$ do
 $A = A+B$

L1:
if $A < C$ goto L2
goto LEND

L2:
if $B > 0$ goto L3
goto LEND

L3: if $A = 1$ goto L4
goto L5

L4:
 ~~$t1 = C+1$~~
 $C = t1$
goto L1

L5: if $A <= 0$ goto L6
goto L1

L6: $t2 = A+B$
 $A = t2$
goto L5

LEND:

(4) generate the three address code for following Program fragment where $(A < C \ \& \ B < D)$ do

if $A = 1$ then $C = C + 1$
else while $A < = 0$ do
 $A = A + B$

L1:
if $A < C$ goto L2
goto LEND

L2:
if $B > 0$ goto L3
goto LEND

L3: if $A == 1$ goto L4
goto L5

L4:
 $t1 = C + 1$
 $C = t1$
goto L1

L5: if $A < = 0$ goto L6
goto L1

L6: $t2 = A + B$
 $A = t2$
goto L5

LEND: